Losing iron sites or losing the nickel host? Why Ni–Fe oxygen-evolution catalysts deactivate

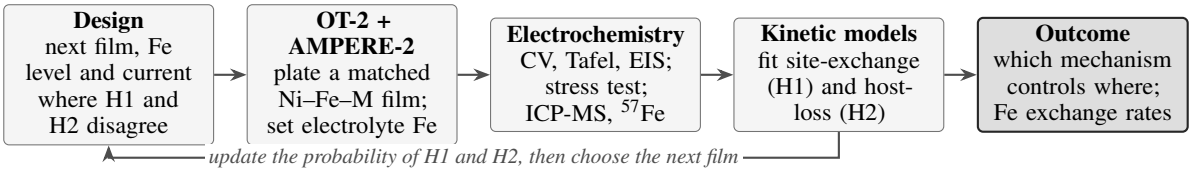
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Research question. Ni–Fe (oxy)hydroxides are among the most active earth-abundant catalysts for the oxygen evolution reaction (OER) in alkaline media [1,2], the anode chemistry of alkaline and anion-exchange-membrane electrolyzers, which need no iridium. Their activity fades under load. Is that because Fe active sites leave faster than they return, or because the Ni host itself transforms?

Knowledge gap. Fe in these catalysts is not fixed: it dissolves and redeposits during OER, and activity holds only while the two rates balance [3]. Chung et al. showed this across several hosts and proposed a descriptor for Fe retention [3]. Whether that exchange, or restructuring of the host such as Ni/Fe segregation, sets durability when both are measured as competing rate laws on the same films has not been tested, because each test consumes a film and needs trace electrolyte analysis.

Hypotheses. *H1, site exchange:* retained activity is set by the steady-state Fe site coverage, fixed by a composition-dependent Fe dissolution rate and an electrolyte-Fe-dependent redeposition rate. Predictions: adding Fe to the electrolyte restores activity, and Fe lost to the electrolyte predicts the activity decay of every composition with the same two rate laws. *H2, host restructuring:* activity loss follows irreversible change in the Ni host (fewer redox-active Ni sites, rising charge-transfer resistance, loss of film). Predictions: added Fe does not restore activity, and decay tracks Ni lost from the film (digested after the test) and a change of phase, not the Fe flux.

Approach. An Opentrons OT-2 fitted with the open AMPERE-2 electrodeposition hardware [4] plates matched Ni–Fe–M films (Fe 0–25%; M = Co or Mn, 0–20%) on identical substrates by one fixed protocol, then conditions them anodically: the films are a controlled variable, not the product. Each film gets CV, a Tafel scan and EIS, then a short stress test (constant current or potential cycling) in Fe-free KOH [2] with Fe added at set levels, while the robot draws electrolyte aliquots for ICP-MS (Fe, Ni, M); an ^{57}Fe spike separates redeposition from dissolution. Both mechanisms are written as kinetic models with shared parameters. Bayesian experimental design picks the next composition, electrolyte Fe level and current density where the models' predicted activity retention differs most, so AI serves mechanism identification.



Expected outcomes. (1) A map of where in composition and electrolyte space each mechanism controls durability; (2) Fe dissolution and redeposition rate constants as functions of host composition; (3) whether a third metal works by slowing Fe loss or by stabilizing the host. Together these explain how to keep earth-abundant active sites in place over an electrolyzer's life.

Fit and resources. The project addresses catalyst *dynamics during catalytic cycles and deactivation*, *mechanisms mediated by earth-abundant metals*, and *AI/ML for mechanism identification*. It does not optimize catalyst synthesis or electrolyzer design. The PI works on self-driving labs and Bayesian experimental design; the lab has an OT-2, is adding a potentiostat, and has campus ICP access.

References. [1] M. W. Louie, A. T. Bell, *J. Am. Chem. Soc.* 135, 12329 (2013), doi:10.1021/ja405351s. [2] L. Trotochaud *et al.*, *J. Am. Chem. Soc.* 136, 6744 (2014), doi:10.1021/ja502379c. [3] D. Y. Chung *et al.*, *Nat. Energy* 5, 222 (2020), doi:10.1038/s41560-020-0576-y. [4] Fisker-Bødker *et al.*, *Digital Discovery* 4, 2491 (2025), doi:10.1039/D5DD00180C.